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Full Length Article

DA-S^{2}RL: Disease-aware shared-specific representation learning for heterogeneous image-EHR Alzheimer's disease diagnosis

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ABSTRACT

Combining electronic health records (EHR) with medical images is important for improving the diagnosis of Alzheimer's disease (AD). However, neuroimages and EHR differ greatly in semantic content and data distribution. Neuroimages mainly reflect brain structure and metabolic abnormalities, while EHR records clinical symptoms and behavioral changes. Directly fusing these heterogeneous modalities may mix shared disease information with modality-specific diagnostic cues, resulting in redundant and ambiguous representations. To address this problem, we propose DA-S^{2}RL, a disease-aware shared-specific multimodal representation learning framework for AD diagnosis. DA-S^{2}RL organizes heterogeneous image and EHR features into disease-related shared semantics and modality-specific diagnostic residuals. The core of DA-S^{2}RL is the shared-specific cross-modal coding (SCC) module. It strengthens cross-modal semantic consistency, suppresses redundant correlations, and enables mutual refinement between shared semantics and diagnostic residuals. Before SCC, image features are integrated into a compact image representation, and EHR features are encoded into the same feature space. Experiments on ADNI and NACC demonstrate that explicitly organizing heterogeneous image-EHR representations into shared disease semantics and modality-specific residuals improves multimodal diagnostic learning under different image-EHR settings.

1. Introduction

Alzheimer's disease is the most common neurodegenerative disease, accounting for about 60–70% of dementia cases (Holtzman et al., 2011). It is mainly manifested by memory loss, cognitive impairment and behavioral abnormalities. In recent years, with the aging of the global population, the number of people suffering from AD has shown a rapid growth trend, and it is expected that by 2050 there will be more than 150 million people suffering from dementia globally (GBD 2019 Dementia Forecasting Collaborators, 2022), which will bring a heavy burden to the social healthcare system. Although there is no effective cure for AD, studies have shown that early identification and intervention can help to slow down the progression of the disease (Rasmussen & Langerman, 2019), so it is important to build an efficient and reliable early diagnosis method.

Although multimodal learning methods have made significant progress in natural image processing (Baltrušaitis et al., 2018), their application to early AD diagnosis remains challenging due to the heterogeneous nature of clinical data. Existing imaging-based approaches often rely on a single modality or adopt shallow concatenation strategies (Kui et al., 2026), which are insufficient to capture the complementary disease information distributed across MRI, PET, and EHR. Such strategies treat multimodal features as independent signals and fail to model their deep semantic dependencies. This limits diagnostic accuracy, particularly in the clinically challenging MCI vs. CN discrimination, where subtle pathological differences require richer cross-modal reasoning. Beyond imaging, EHR data, encompassing structured clinical symptoms, cognitive assessments, and behavioral observations, provide complementary evidence of neurodegeneration that is not directly observable in neuroimaging alone. However, integrating EHR with imaging modalities introduces significant scale and semantic disparities: imaging features are spatially dense and continuous, while EHR features are sparse and heterogeneous. More importantly, imaging and EHR describe AD from different views. Neuroimaging mainly reflects structural and metabolic abnormalities, whereas EHR records symptom-level and behavioral manifestations. Therefore, imaging-EHR fusion is not only a feature fusion problem, but also a shared-specific representation learning problem. Some studies have attempted unified fusion frameworks to address this gap; for example, Duenias et al. (2025) proposed HyperFusion, which achieves dynamic cross-modal synergy via

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hyper-network modeling, and Prabhu et al. (2022) employed separate sub-networks for MRI and EHR feature extraction. However, these methods often rely on coarse-grained fusion and lack explicit modeling of cross-modal common evidence and modality-specific diagnostic cues. As a result, disease-related evidence shared by different modalities and useful modality-dependent cues may be mixed in a unified feature space, leading to redundant or ambiguous representations and limiting the use of complementary diagnostic information.

To address the aforementioned challenges, this paper proposes DA-S^{2}RL, a disease-aware shared-specific diagnostic framework for Alzheimer's disease (AD) diagnosis. DA-S^{2}RL primarily organizes representations based on their diagnostic value, adapting shared-specific representation learning to heterogeneous imaging-EHR diagnostic tasks. Traditional shared-private or decoupling methods mainly separate cross-modal common information from modality-specific information; DA-S^{2}RL further distinguishes between two types of information: disease consistency evidence supported by both imaging and EHR, and modality-specific diagnostic residuals that still possess diagnostic value after suppressing shared associations. Based on this model architecture, the SCC module collaboratively implements cross-modal semantic consistency constraints, redundant information suppression, and bidirectional shared-residual optimization, thereby simultaneously preserving diagnostic evidence with both commonalities and complementarities. The main contributions of this paper are as follows:

- **Disease-oriented shared-specific representation framework for imaging-EHR diagnosis:** We formulate heterogeneous imaging-EHR fusion from the perspective of diagnostic roles. Unlike conventional shared-private or disentangled multimodal methods that primarily decompose representations into cross-modal common information and modality-dependent information, DA-S^{2}RL further organizes them into cross-modal shared diagnostic evidence and complementary modality-dependent diagnostic residuals, thereby modeling information jointly supported by different modalities as well as modality-specific information that remains diagnostically informative beyond the shared component.
- **Shared-specific cross-modal coding for heterogeneous multimodal fusion:** We design SCC to organize imaging-EHR representations through cross-modal semantic consistency, redundancy suppression, and CDAB-based bidirectional refinement. Contrastive consistency encourages disease-related correspondence between the imaging and EHR-side shared representations. Mutual-information-based regularization reduces redundant dependencies between shared semantics and diagnostic residuals. CDAB further refines the bidirectional interaction between shared representations and modality-specific diagnostic residuals.
- **EHR analysis across AD-related classification tasks:** We group the 28 discrete EHR symptom features for ADNI into clinically relevant categories and use SHAP analysis to examine their task-specific diagnostic contributions. Experiments on ADNI and NACC further demonstrate the effectiveness of DA-S^{2}RL in imaging-EHR diagnostic under different modality settings.

2. Related work

Early studies on Alzheimer's Disease (AD) classification mainly focused on single-modality neuroimaging data such as structural MRI or functional PET, and were usually achieved through manually extracted features or convolutional neural network (CNN) modeling of brain morphological or functional features (Payan & Montana, 2015). These methods have laid the foundation of image-assisted diagnosis, but their generalization ability and robustness are still limited due to the difficulty of capturing the heterogeneous pathological manifestations of AD comprehensively in a single modality.

In terms of deep modeling, Convolutional Neural Network (CNN) structures are widely used in image data to capture spatial local and hierarchical features (Erdogmus & Kabakus, 2023). Although these methods have achieved good results in unimodal modeling, it is still difficult for a single modality to comprehensively reflect the complex pathological process of AD development due to the significant differences in the structural and functional information carried by different modalities. Therefore, researchers have begun to focus on multimodal fusion strategies for MRI and PET. Most of the existing multimodal methods use a simple two-branch convolutional network structure (Kong et al., 2022), and achieve feature fusion through concatenation or attention mechanisms. In addition, Choudhury et al. (2024) proposed a CGAN framework for MRI and PET fusion, showing good performance in AD classification. However, such methods usually focus on shallow fusion and fail to adequately model cross-modal commonality and modality-dependent complementarity. Some studies have further introduced the Transformer architecture (Duong et al., 2025; Zhao et al., 2026). Although this approach improves the performance of modal fusion, it still suffers from modal redundancy (Jia et al., 2026). Recently, He et al. (2025) proposed a flexible optimal graph fusion method for multimodal low-rank sparse features and achieved competitive performance in AD detection. However, this method may limit its ability to fully exploit complementary information across modalities.

To better model cross-modal complementarity, recent studies have explored shared-specific representation learning, often combined with attention mechanisms to enhance semantic correspondence (Sanchez et al., 2020). Existing shared-private or disentangled multimodal formulations generally decompose representations into cross-modal common information and modality-dependent information (Hu et al., 2025; Liu et al., 2018; Shen et al., 2025). However, for heterogeneous imaging-EHR diagnosis, modality-dependent information may still contain complementary diagnostic evidence beyond that captured by the shared component. This motivates organizing shared-specific representations according to their diagnostic roles, rather than only according to cross-modal commonality and modality dependence. Such shared-specific modeling remains underexplored in AD classification, particularly for heterogeneous imaging-EHR integration. Concurrently, electronic health records (EHR) are increasingly integrated into AD research (Esmaeilzadeh et al., 2018; Nie et al., 2026). While early methods relied on simple feature concatenation (Venugopalan et al., 2021), recent works employ tabular attention or multimodal Transformers to capture semantic relationships (Liu et al., 2023b). However, these approaches still provide limited modeling of shared disease evidence and modality-dependent diagnostic cues.

Motivated by these limitations, we propose DA-S^{2}RL, a disease-aware shared-specific multimodal representation learning framework for AD diagnosis. DA-S^{2}RL organizes heterogeneous imaging-EHR representations into cross-modal shared diagnostic evidence and complementary modality-dependent diagnostic residuals. SCC implements this organization through cross-modal consistency learning, redundancy reduction, and bidirectional shared-specific refinement, while providing interpretable evidence from clinical features.

3. Proposed method

A multimodal diagnostic framework is presented for AD classification by integrating imaging and EHR data. The proposed framework follows a heterogeneity-aware shared-specific representation learning strategy, which assigns heterogeneous multimodal information into cross-modal shared diagnostic evidence and modality-dependent diagnostic residuals. Imaging features are first integrated into a compact imaging representation, while EHR features are encoded into an aligned feature space. The central component is SCC, which organizes imaging-EHR representations into these two diagnostic roles: the shared representation captures diagnostic information jointly supported by both modalities, whereas the residual representations preserve complementary clinical or imaging cues that cannot be fully explained by the

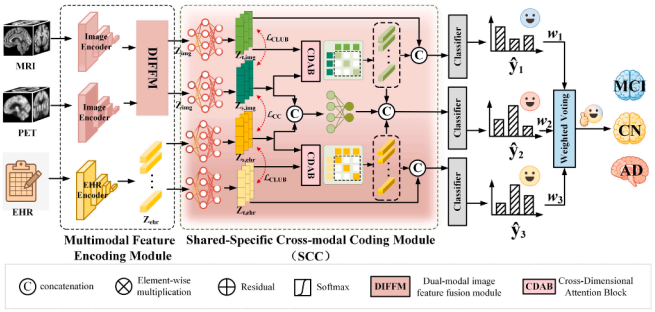


Fig. 1. Overview of the proposed DA-S^{2}RL framework. MRI, PET, and EHR data are first encoded separately. MRI and PET features are integrated by DIFFM to obtain a compact imaging representation, which is then combined with the EHR representation through the Shared-Specific Cross-modal Coding (SCC) module. SCC organizes heterogeneous imaging-EHR representations into disease-related shared semantics and modality-specific diagnostic residuals, followed by branch-level prediction heads for final decision fusion.

shared semantics. Branch-level prediction heads are then used for final decision fusion. The overall framework is shown in Fig. 1.

3.1. Multimodal feature encoding module

To fully extract the discriminative information embedded in multimodal data, this study designs independent feature encoding modules for MRI, PET, and structured tabular data. This design aims to preserve the original feature distributions and semantic structures of each modality, while providing unified and discriminative representations for subsequent modality fusion and classification tasks.

3.1.1. Imaging feature encoding

MRI and PET volumes are processed through two parallel 3D ResNet branches to extract modality-specific spatial features. Given input volumes $I_{\mathrm{MRI}} \in \mathbb{R}^{D\times H\times W}$ and $I_{\mathrm{PET}} \in \mathbb{R}^{D\times H\times W}$, these two encoders operate independently with non-shared weights to accommodate modality-specific distributional differences. The 3D ResNet18 architecture used is initialized with pretrained weights from Tencent Medical AI, derived from 23 publicly available medical imaging datasets, which enhances both generalization and transfer capabilities.

The encoder architecture is based on the ResNet3D design and consists of four downsampling stages, each composed of multiple 3D convolutional layers, BatchNorm3D, and ReLU activation functions. This structure enables progressive extraction of high-level semantic features and effective nonlinear modeling. The input to the network is a single-channel 3D tensor (i.e., MRI or PET volume), which is processed through the convolutional backbone, followed by adaptive 3D average pooling, and finally mapped to a 512-dimensional embedding space via a fully connected layer. This strategy enables efficient representation learning from raw imaging data while leveraging pretrained knowledge.

3.1.2. EHR feature encoding

The structured EHR features contain both continuous and discrete attributes. We first preprocess each attribute according to its type. Continuous attributes are directly retained after normalization, while discrete attributes are converted into one-hot vectors according to their category cardinalities. For the ith feature segment v_i with category cardinality L_i, the preprocessing is defined as:

$$\mathbf{z}_i = \begin{cases} v_i, & \text{if } L_i = 1, \\ \mathrm{OneHot}\big(\mathrm{clamp}(\mathrm{round}(v_i), 0, L_i - 1)\big), & \text{if } L_i > 1. \end{cases} \tag{1}$$

The processed EHR vector is obtained by concatenating all feature segments:

$$\mathbf{z}_{\mathrm{ehr}} = \mathrm{Concat}(\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_M). \tag{2}$$

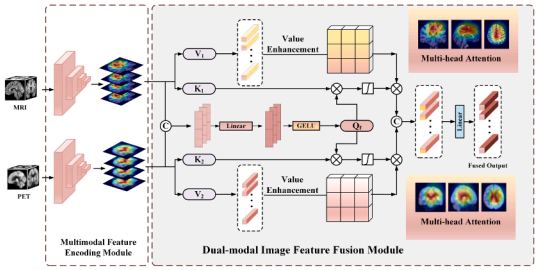


Fig. 2. The architecture of the dual-modal image feature fusion module (DIFFM) for MRI and PET, where the network outputs a fused representation as a vector of shape (B, 512).

The preprocessed EHR vector is then encoded by a Transformer-based tabular encoder and projected into a 512-dimensional representation:

$$F_{\mathrm{ehr}} = E_{\mathrm{ehr}}(\mathbf{z}_{\mathrm{ehr}}). \tag{3}$$

The resulting F_{ehr} is used as the EHR-side input for the subsequent Shared-Specific Cross-modal Coding (SCC) module.

3.2. Dual-modal image feature fusion module

Before modeling the heterogeneity between imaging and EHR features, MRI and PET representations are first integrated into a compact imaging feature. This step provides a unified imaging input for the subsequent Shared-Specific Cross-modal Coding (SCC) module. We implement this process using a lightweight Dual-modal Image Feature Fusion Module (DIFFM), as shown in Fig. 2.

Given MRI and PET embeddings, DIFFM derives a joint query from their concatenation to guide cross-modal feature aggregation. To introduce lightweight dimension-wise interaction, the value features are processed by a reshape-convolution-flatten operation. Specifically, a value feature $x \in \mathbb{R}^{B\times 512}$ is reshaped into $\mathbb{R}^{B\times 1\times 16\times 32}$ before the 2D convolution:

$$\Phi_{\mathrm{VME}}(x) = x + \mathrm{Flatten}(\mathrm{Conv2d}(\mathrm{Reshape}(x))). \tag{4}$$

Here, Reshape$(\cdot)$ reorganizes the 512-dimensional feature vector into a 16×32 structured layout, while Flatten$(\cdot)$ restores the convolution output to $\mathbb{R}^{B\times 512}$ for residual addition. Φ_{VME} is used to enhance local dependencies among feature dimensions within the reorganized layout.

In the fusion stage, the Query (Q) is generated from the concatenated MRI and PET features, while the Keys (K) are obtained through linear projections. The Values (V) are processed by Φ_{VME} before aggregation. The projected Q, K, and V are 512-dimensional representations. We use 8 attention heads, with each head having a dimension of $d_k = 64$. A sigmoid-based gating operation is then independently applied within each attention head to weight the enhanced values:

$$\alpha = \sigma(QK^T/\sqrt{d_k}). \tag{5}$$

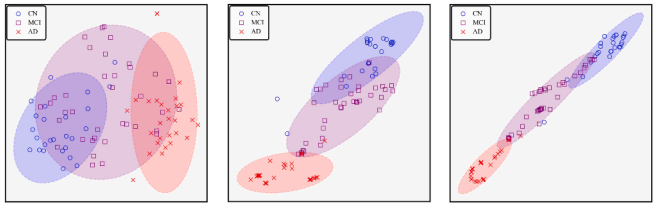
The fused imaging representation is obtained as:

$$F_{\mathrm{fused}} = \mathrm{Proj}(\mathrm{Concat}(\alpha_1 V_1,\ \alpha_2 V_2)). \tag{6}$$

The resulting F_{fused} serves as the imaging-side representation for SCC, where it is further modeled together with the EHR representation through shared-specific cross-modal coding.

3.3. Shared-specific cross-modal coding module

To reduce ambiguous feature mixing in imaging-EHR fusion, we propose the Shared-Specific Cross-modal Coding (SCC) module. SCC provides a disease-aware organization of heterogeneous diagnostic evidence. In this module, the shared subspace is encouraged to encode disease-consistent semantics supported by both imaging and EHR data.



(a) Before SCC (b) After SCC (shared) (c) After SCC (specific)

Fig. 3. t-SNE visualization of feature distributions before and after SCC. (a) shows the feature distribution before SCC, where the classes overlap. (b) and (c) show the learned shared and modality-dependent representations after SCC, respectively, indicating improved class separation.

The modality-specific subspaces are not treated as generic private features; instead, they are formulated as diagnostic residuals, which refer to complementary modality-dependent cues that remain informative after shared disease semantics have been extracted and redundant dependencies have been suppressed. In this way, SCC allows the model to jointly use cross-modal common evidence and modality-unique clinical cues. The effectiveness of the SCC module is illustrated in Fig. 3. The t-SNE visualization shows the feature distribution before and after SCC. In (a), the classes overlap before SCC, with a silhouette score of 0.208. After SCC, the shared and modality-dependent features in (b) and (c) achieve higher scores of 0.461 and 0.504, respectively, indicating clearer inter-class separation and more discriminative representations. Specifically, given the fused image features F_{fused} from DIFFM and EHR features F_{ehr} from the encoder, we apply parallel MLP heads to project each modality into a disease-consistent shared semantics z_s and a modality-specific representation z_t, where $Z \in \{F_{\text{fused}}, F_{\text{ehr}}\}$. To reduce redundant dependencies between z_s and z_t, we employ the Contrastive Log-ratio Upper Bound (CLUB) estimator to minimize an upper bound of mutual information, defined as:

$$\mathcal{L}_{\text{CLUB}}(z_s, z_t) = \mathbb{E}_{(z_s,z_t)}\left[-\frac{1}{2}\|\mu(z_s) - z_t\|^2\right] + \mathbb{E}_{(z_s,z_t')}\left[\frac{1}{2}\|\mu(z_s) - z_t'\|^2\right], \tag{7}$$

where $\mu(z_s)$ is the predicted mean of z_t using a learned MLP, and z_t' denotes a randomly permuted sample. To introduce disease-level supervision into the shared subspace, we employ a disease-conditioned cross-modal contrastive consistency loss between the imaging-shared and EHR-shared representations. The image-shared features $z_{s,\text{img}}$ and EHR-shared features $z_{s,\text{ehr}}$ are passed through projection heads g_i and g_t to compute the following loss:

$$\mathcal{L}_{\text{CC}} = -\frac{1}{B}\sum_{i=1}^{B}\frac{1}{|P(i)|}\sum_{p\in P(i)}\log \frac{\exp\left(\text{sim}\left(g_i(z_{s,\text{img}}^{(i)}), g_t(z_{s,\text{ehr}}^{(p)})\right)/\tau\right)}{\sum_{a=1}^{B}\exp\left(\text{sim}\left(g_i(z_{s,\text{img}}^{(i)}), g_t(z_{s,\text{ehr}}^{(a)})\right)/\tau\right)}. \tag{8}$$

Here, y_i denotes the diagnostic label of the ith subject, and $P(i) = \{p \mid y_p = y_i\}$ denotes the set of EHR samples sharing the same diagnostic label as the ith imaging anchor, including the paired EHR sample from the same subject. Therefore, both the paired cross-modal sample and samples from other subjects with the same diagnosis are treated as positives, whereas samples from different diagnostic categories serve as contrastive negatives. τ is the temperature parameter, and $\text{sim}(\cdot,\cdot)$ denotes cosine similarity between normalized projected representations. As the final refinement component of SCC, we introduce a Cross-Dimensional Attention Block (CDAB), as shown in Fig. 4, to refine the learned shared and modality-specific representations. CDAB is designed to model the interaction between common disease-related semantics and modality-dependent diagnostic cues through two complementary attention paths, namely shared-to-specific and specific-to-shared refinement:

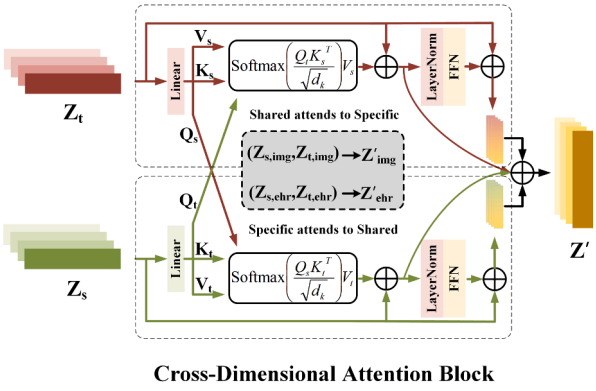


Fig. 4. Bidirectional shared-specific refinement within SCC.

$$z'_{\text{img}} = \text{CDAB}(z_{s,\text{img}}, z_{t,\text{img}}), \quad z'_{\text{ehr}} = \text{CDAB}(z_{s,\text{ehr}}, z_{t,\text{ehr}}). \tag{9}$$

The aligned shared representations $z_{s,\text{img}}$ and $z_{s,\text{ehr}}$ are then concatenated and projected to obtain the unified global shared vector $z_s = \text{Linear}(\text{Concat}(z_{s,\text{img}}, z_{s,\text{ehr}}))$. We concatenate the original and CDAB-enhanced modality-dependent features for each modality to obtain:

$$z_{\text{img}}^{\text{final}} = \text{Concat}(z_{t,\text{img}}, z'_{\text{img}}), \quad z_{\text{ehr}}^{\text{final}} = \text{Concat}(z_{t,\text{ehr}}, z'_{\text{ehr}}). \tag{10}$$

The total training loss for the SCC module is defined as:

$$\mathcal{L}_{\text{SCC}} = \lambda_{\text{CC}} \cdot \mathcal{L}_{\text{CC}} + \lambda_{\text{CLUB}} \cdot \left(\mathcal{L}_{\text{CLUB}}^{\text{img}} + \mathcal{L}_{\text{CLUB}}^{\text{ehr}}\right), \tag{11}$$

where λ_{CC} and λ_{CLUB} are scalar hyperparameters. In this way, CDAB complements the contrastive alignment and mutual-information regularization in SCC by explicitly enabling bidirectional interaction between shared and modality-specific representations.

3.4. Prediction and optimization

To fully leverage the shared-specific and cross-modality-enhanced representations from the SCC module, a multi-branch classification strategy is designed. Building on the representations defined in Section 3.3, the three branch inputs are organized as follows: the image branch takes $z_{\text{img}}^{\text{final}}$, the tabular branch takes $z_{\text{ehr}}^{\text{final}}$, and the fused branch takes $\text{Concat}(z_s, z'_{\text{img}}, z'_{\text{ehr}})$, which integrates the global shared representation with the CDAB-enhanced outputs from both modalities, each encoding the interaction between shared and modality-specific features.

Based on this, the proposed framework constructs three parallel prediction branches, namely the image branch, fusion branch, and EHR branch. Each branch is fed into a lightweight prediction head to obtain branch-level outputs $\hat{y}_1$, $\hat{y}_2$, and $\hat{y}_3$. This design preserves the diagnostic contributions of different representation types after SCC, while avoiding additional cross-modal mixing before decision making. The final prediction is obtained through decision-level weighted fusion.

$$\hat{y} = \sum_{i=1}^{3}\tilde{w}_i\hat{y}_i, \quad \tilde{w}_i = \frac{w_i}{\sum_{j=1}^{3} w_j}, \quad \sum_{i=1}^{3}\tilde{w}_i = 1. \tag{12}$$

where w_i denotes the voting weight assigned to the ith prediction branch, and $\tilde{w}_i$ is its normalized form. To further enhance the supervision of fused image features, an early classification head is introduced in the DIFFM module to independently predict labels from the fused feature F_{fused}, and an auxiliary loss is computed accordingly:

$$\mathcal{L}_{\text{fused-cls}} = \text{CrossEntropy}(\hat{y}_{\text{fused}}, y_{\text{true}}). \tag{13}$$

Although this branch does not contribute to final inference, it provides additional gradients during training to strengthen discriminative learn-

Table 1
Demographic information of subjects from ADNI and NACC datasets, along with EHR feature composition and variables.

Dataset	Group	Number	Gender (M/F)	Age (Mean ± Std)	EHR Features Composition	EHR Variables
ADNI	CN	101	51/50	77.11±5.93	28 discrete features	discrete features: NAUSEA, VOMIT, DIARRH, CONSTP, ABDOMN, SWEATN, DIZZY, ENERGY, DROWSY, VISION, HDACHE, DRYMTH, BREATH, COUGH, PALPIT, CHEST, URNDIS, URNFRQ, ANKLE, MUSCLE, RASH, INSOMN, DPMOOD, CRYING, ELMOOD, WANDER, FALL, OTHER
	MCI	118	61/57	77.61±7.81		
	AD	81	49/32	75.81±5.60		
NACC	CN	260	65/195	68.82±8.18	18 discrete and 6 continuous features	discrete features: DEPD, AFIBRILL, CONGHRT, SEX, HEARAID, BOWLING, ENERGY, BILLS, TAXES, SHOPPING, GAMES, STOVE, MEALPREP, EVENTS, PAYATTN, REMDATES, TRAVEL, MOT; continuous features: MOCATOTS, TRAILA, NACCAGE, EDUC, ANIMALS, NACCGDS
	MCI	163	72/91	72.33±7.62		
	AD	227	104/123	70.93±9.48		

ing in early fusion stages. The final training loss integrates all components as follows:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{SCC}} + \lambda_{\text{fused-cls}} \cdot \mathcal{L}_{\text{fused-cls}}, \tag{14}$$

where $\mathcal{L}_{\text{CE}}$ represents the summed cross-entropy losses from the three classification branches, and $\lambda_{\text{fused-cls}}$ is a weighting hyperparameter.

4. Experiments

4.1. Datasets and pre-processing

This study primarily utilizes the multimodal dataset from the Alzheimer's Disease Neuroimaging Initiative (ADNI) for modeling and analysis (Mueller et al., 2005). ADNI is a large-scale clinical program covering multicenter, longitudinal follow-up, and is widely used in studies of early diagnosis and progression of AD (Petersen et al., 2010). The dataset includes rich structural MRI, PET, EHR, genetic, and neuropsychological assessment data. To achieve temporal alignment across modalities, we adopt a timestamp-based pairing strategy with a predefined 6-month matching window.

To demonstrate the effectiveness of our proposed method, we also curated the National Alzheimer's Coordinating Center (NACC) dataset for auxiliary validation. Due to the limited availability of PET scans in the NACC dataset, we employed a multimodal MRI pairing strategy, using MPRAGE and T2FLAIR sequences. These MRI images were integrated with EHR data to construct aligned multimodal samples. Subjects were included if they had a valid diagnostic label, the required imaging modalities, and an eligible complete EHR record within the predefined temporal window. Subjects were excluded if any required modality was unavailable, the diagnostic label was missing or ambiguous, or no temporally matched eligible visit could be identified. Subjects with missing values in any of the selected EHR variables were excluded, and no additional missing-value imputation was performed. For ADNI, diagnostic labels were assigned according to the diagnostic categories provided in the downloaded data (AD, MCI, and CN). For NACC, diagnostic labels were determined using the NACCUDSD and NACCALZD fields: NACCUDSD = 1 was defined as CN, NACCUDSD = 3 as MCI, and subjects satisfying both NACCUDSD = 4 and NACCALZD = 1 were defined as AD. When multiple eligible records were available for a subject, the record with the smallest temporal interval between the imaging and EHR data was selected, and only one multimodal sample was retained for each subject. The numbers of subjects retained after major cohort filtering steps are summarized in Table 3.

Therefore, ADNI is used to evaluate the MRI-PET-EHR setting, while NACC is used to further examine the adaptability of the proposed imaging-EHR framework under a PET-unavailable scenario. Table 1 summarizes the demographic characteristics and complete EHR variables of both datasets. Variables directly used to determine the diagnostic labels were not included as model inputs to avoid direct label leakage.

In terms of image data preprocessing, a set of standardized processing procedures was adopted in this study. First, the FSL tool was used for skull stripping to remove non-brain tissue regions and enhance the visual representation of brain structures. Subsequently, the N4ITK (Tustison et al., 2010) algorithm was applied for bias field correction of the MRI images, aiming to correct the intensity inhomogeneity artifact problem due to magnetic field non-uniformity. In the image alignment stage, ANTs (Avants et al., 2014) were used for spatial registration between paired imaging modalities (MRI–PET for ADNI and MPRAGE–T2FLAIR for NACC). In order to unify the input data dimensions and adapt the network structure, all images were resampled to a spatial size of $182 \times 218 \times 182$ with isotropic voxel size of $1 \times 1 \times 1$ mm^3 at the end of preprocessing.

The tabular data include multiple clinically observed features spanning mood, sleep, behavior, physical status, and autonomic symptoms (e.g., palpitations, dizziness). These semantically distinct indicators are

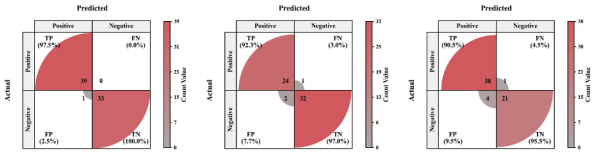
Table 2
Performance comparison with representative methods on ADNI and NACC datasets.

Dataset	Method	MRI	PET	EHR	ACC (%)	AUC (%)
ADNI	HyperFusion (Duenias et al., 2025)	✓	✗	✓	67.30	82.20
	DAFT (Wolf et al., 2022)	✓	✓	✓	73.30	87.40
	Zhang et al. (2025)	✓	✗	✓	78.60	91.30
	3MT (Liu et al., 2023a)	✓	✓	✓	75.80	89.60
	HybridCA-Net (Zhou et al., 2025)	✓	✗	✗	85.64	88.94
	Turrisi and Patané (2026)	✓	✗	✗	74.50	91.40
	Ours (w/o PET)	✓	✗	✓	89.13	89.69
	Ours	✓	✓	✓	**93.48**	**94.39**
NACC	Guillén and Frias-Martinez (2025)	✗	✗	✓	89.50	–
	Yi et al. (2023)	✗	✗	✓	80.52	88.00
	Alatrany et al. (2024)	✗	✗	✓	85.20	–
	Qiu et al. (2022)	✓	✗	✓	80.40	–
	Wang et al. (2024)	✗	✗	✓	90.21	92.13
	Yang et al. (2026)	✗	✗	✓	83.38	93.98
	Ours (w/o Imaging)	✗	✗	✓	89.69	93.16
	Ours	✓	✗	✓	**93.81**	**98.96**

Note: Bold indicates best performance. For NACC, MRI denotes paired multi-sequence MRI, including MPRAGE and T2FLAIR; PET is not included due to limited availability.

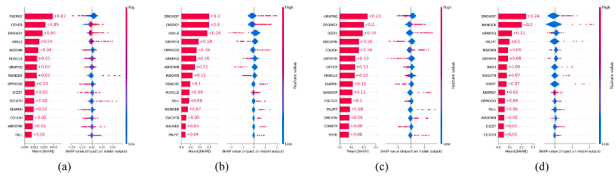
Table 3
Subject counts after major cohort filtering steps for ADNI and NACC.

Filtering step	ADNI	NACC
Required paired imaging available	810	1120
Complete EHR record available	428	771
Image–EHR temporal matching satisfied	300	650



(a) AD VS. CN (b) AD VS. MCI (c) MCI VS. CN

Fig. 5. The confusion matrices for three binary classification tasks on the NACC dataset.



(a) (b) (c) (d)

Fig. 6. SHAP-based visualization of EHR feature contributions across Alzheimer's disease classification tasks. (a) shows the overall feature importance for the multi-class classification task (AD vs. MCI vs. CN). (*b*)–(*d*) present SHAP-based feature contributions for three binary classification tasks: AD vs. CN, AD vs. MCI, and MCI vs. CN.

linked to early cognitive and behavioral impairments in AD. Notably, symptoms such as depressed mood, insomnia, and crying are associated with hippocampal and prefrontal atrophy (Cavaillès et al., 2025), while falling, drowsiness, and blurred vision reflect deficits in motor coordination and spatial perception (Minta et al., 2023).

4.2. Implementation details

The proposed multimodal classification framework is implemented using PyTorch. All imaging volumes are resampled to a unified shape of $182 \times 218 \times 182$. The experiments were conducted on a computer equipped with an NVIDIA L40 GPU. The training objective combines cross-entropy loss, a disease-conditioned cross-modal contrastive consistency loss, CLUB-based mutual information minimization, and an auxiliary classification loss from the fusion stage. The loss weights are empirically set to $\lambda_{\mathrm{CC}} = 0.5$, $\lambda_{\mathrm{CLUB}} = 1.0$, and $\lambda_{\mathrm{fused-cls}} = 0.2$. We use the Adam optimizer with an initial learning rate of 1×10^{-4}, a batch size of 16, and a cosine annealing learning rate schedule with a 10-epoch warm-up. The model is trained for 300 epochs.

4.3. Evaluation metrics

To comprehensively evaluate the performance of the model, we adopt a variety of metrics across both multi-class and binary classification tasks, including Accuracy (ACC), Sensitivity (SEN), Specificity (SPE), and Area Under the ROC Curve (AUC). These metrics collectively reflect the model's capabilities in terms of classification accuracy, discriminative power between positive and negative classes, and overall robustness.

4.4. Results and analysis

The proposed method is evaluated on the ADNI and NACC datasets under their corresponding modality settings. Specifically, ADNI evaluates the MRI-PET-EHR configuration, whereas NACC uses the multi-sequence MRI-EHR configuration due to limited PET availability. Within each diagnostic category, the datasets were randomly split at the subject level into training, validation, and test sets with a ratio of 7:1.5:1.5. Data augmentation was applied only to the training images, including random rotation, random cropping, resizing, and intensity-preserving spatial transformations. In addition to this fixed split used for the main experiments, we conducted five independent runs with different random seeds for controlled baseline comparison and statistical stability analysis.

To comprehensively evaluate the effectiveness of the proposed method, we conducted systematic comparisons with several representative multimodal approaches across two settings: the cognitive assessment (COG) task and additional binary classification tasks. Table 2 summarizes the performance of different methods on the cognitive status classification (COG) task, which is formulated as a three-class classification problem among AD, MCI, and CN. These results suggest that organizing heterogeneous imaging-EHR data into shared disease semantics and modality-specific residual evidence is beneficial for cognitive status recognition. Specifically, the method achieves an accuracy of 93.48% and an AUC of 94.39% on the ADNI dataset, while attaining an accuracy of 93.81% and an AUC of 98.96% on the NACC dataset. To provide a controlled comparison, we re-evaluated representative multimodal fu-

Table 4
Performance comparison on ADNI dataset for three binary classification tasks (%).

Dataset	Method	AD vs. CN				MCI vs. CN				AD vs. MCI			
		ACC	AUC	SEN	SPE	ACC	AUC	SEN	SPE	ACC	AUC	SEN	SPE
ADNI	WMCL-HN (Yu et al., 2025)	93.74	98.05	94.73	93.18	69.91	75.53	72.72	66.81	–	–	–	–
	MMSDL (Abdelaziz et al., 2025)	95.25	94.68	97.35	92.71	85.22	81.36	69.36	**94.00**	–	–	–	–
	PG-BrainBagNet (Park et al., 2023)	85.25	89.03	84.00	87.18	76.00	80.34	74.36	77.78	81.25	88.92	80.00	82.05
	CE-AH (Wang et al., 2026)	94.90	97.30	95.50	94.30	–	–	–	–	–	–	–	–
	MDL-Net (Qiu et al., 2024)	**96.37**	**98.48**	**97.40**	95.38	73.61	76.08	73.01	73.02	85.29	88.16	93.09	72.00
	Ours	96.30	97.25	92.31	**100.00**	**87.88**	**86.09**	**84.21**	92.86	**93.75**	**97.44**	**94.74**	**92.31**

Note: Bold values indicate the best performance for each metric. The binary classification results in this table are obtained from separately trained pairwise binary models for each diagnostic task.

Table 5
Performance comparison with representative multimodal methods over five random seeds.

Dataset	Method	ACC (%)	AUC (%)
ADNI	LMF	68.70±6.26[b]	84.10±4.90[b]
	MISA	80.00±4.18[b]	91.28±1.88[a]
	ITCFN	67.39±7.99[c]	86.97±4.55[b]
	TriLightNet	77.39±3.30[c]	92.06±1.09[a]
	Ours	**91.74±1.82**	**95.07±3.11**
NACC	LMF	75.92±7.99[b]	86.39±5.92[b]
	MISA	86.73±1.25[a]	96.14±0.41[a]
	ITCFN	78.16±6.79[b]	91.77±2.69[b]
	TriLightNet	78.16±5.03[b]	92.88±2.51[a]
	Ours	**91.34±2.01**	**97.63±1.17**

Note: Results are reported as mean ± standard deviation over five independent runs with different random seeds. Statistical significance was assessed using a two-sided paired t-test between each baseline and our method.

[a] $p < 0.05$,
[b] $p < 0.01$,
[c] $p < 0.001$.

sion methods and shared-private representation learning baselines using the same subject-level data split (Hazarika et al., 2020; Hu et al., 2025; Liu et al., 2018; Shen et al., 2025), preprocessing pipeline, and input modalities as DA-S^{2}RL. Different from the main evaluation in Table 2, which reports results on a fixed subject-level split, Table 5 evaluates robustness under five independent random seeds. Paired significance tests were further performed between each baseline and DA-S^{2}RL across the five runs. DA-S^{2}RL achieves the highest mean ACC and AUC on both ADNI and NACC, with statistically significant differences from the evaluated baselines.

Furthermore, Table 4 presents the results on the ADNI dataset for three additional binary classification tasks, reporting four key metrics – accuracy, sensitivity, specificity, and AUC – to comprehensively assess the diagnostic performance against available comparison methods. Since directly comparable binary-task baselines are limited for the NACC dataset, we additionally report the confusion matrices of our method on three NACC binary classification tasks in Fig. 5.

To further illustrate the effect of decision-level fusion, we visualize the prediction-output distributions of the single fusion branch and the three-branch weighted fusion strategy on the ADNI binary classification tasks, as shown in Fig. 7. Specifically, the prediction scores from the fusion branch $\hat{y}_2$ and the final weighted prediction $\hat{y}$ are visualized using t-SNE. Compared with the single fusion branch, the weighted fusion strategy exhibits clearer separation between diagnostic categories in the decision-output space, suggesting that combining complementary predictions from different branches improves decision-level discrimination. Overall, experiments on the ADNI and NACC datasets confirm that the proposed DA-S^{2}RL achieves competitive performance on the COG task and binary classification benchmarks, with structured EHR features and the shared-specific representation mechanism both contributing to the observed improvements.

To investigate the contributions of different EHR feature groups on the ADNI dataset, which comprises 28 discrete clinical symptom features, we conducted a subgroup ablation study by progressively organizing three clinically defined groups into the imaging baseline (Table 6): G3 Neuropsychiatric (INSOMN, WANDER, DPMOOD, CRYING, ELMOOD), G2 Autonomic nervous (DIZZY, ENERGY, DROWSY, DRYMTH, VISION, FALL), and G1 Somatic (gastrointestinal, cardiopulmonary, urinary, musculoskeletal symptoms). The results show a generally consistent improvement trend as more evidence groups are incorporated. Including G3 alone improves three-class accuracy from 86.96% to 89.13% and AD vs. MCI AUC from 94.33% to 94.74%, suggesting that behavioral and sleep-related symptoms contribute stable diagnostic evidence across disease stages. Further incorporation of G2 advances accuracy to 91.30% and AD vs. MCI AUC to 95.95%, reflecting the role of autonomic dysfunction in distinguishing the MCI-to-AD transition. The inclusion of G1 further enhances performance, suggesting that somatic symptoms provide supplementary diagnostic evidence when combined with other features, though their standalone pathological specificity may be weaker. To further understand feature-level contributions, we applied SHAP analysis on the ADNI dataset across all tasks (Fig. 6, Table 7). The results reveal distinct task-specific patterns: autonomic features dominate AD vs. CN discrimination, somatic and autonomic signals are more prominent in AD vs. MCI, and behavioral features such as WANDER emerge relatively early in MCI vs. CN. These patterns suggest that different EHR feature groups contribute complementary diagnostic evidence, which is preserved in the modality-specific diagnostic residuals. This further highlights that EHR-specific residuals carry valuable symptom-level information, enhancing the model's ability to capture subtle disease variations across stages. Detailed feature-level interpretations are provided in Table 7.

4.5. Ablation experiments

To examine the roles of shared and modality-specific representations, we conduct a representation-level ablation study (Table 9). AUC is used as the primary metric to assess the discriminative contribution of different representation types. M1, M2, and M3 represent the shared-only, modality-specific-only, and joint shared-specific settings, respectively, while M4 denotes the full model. To avoid the influence of decision-level fusion, M1–M3 use a single classification head without weighted voting. M3 achieves a higher AUC than M1 and M2, improving from 89.75% and 88.95% to 91.17%, showing that shared and modality-specific representations provide complementary diagnostic information. The full model further improves AUC to 94.39%, demonstrating the benefit of branch-level prediction and weighted voting.

All analyzes in Tables 8 and 11 use frozen representations and follow the same data split as the main experiments. For the probing analyzes in Table 8, linear probes are fitted using representations from the training subjects and evaluated on the held-out test subjects without updating the representation encoders. For cross-modal

Table 6
EHR subgroup ablation study on ADNI dataset (%).

Config	Configuration	COG task		AD vs. MCI			
		ACC	AUC	ACC	AUC	SEN	SPE
0	Imaging only	86.96	88.90	84.38	94.33	76.92	89.47
1	Imaging + G3	89.13	90.61	84.38	94.74	69.23	94.74
2	Imaging + G3 + G2	91.30	92.21	87.50	95.95	76.92	94.74
3	Imaging + G3 + G2 + G1	**93.48**	**94.39**	**90.62**	**96.36**	**84.62**	**94.74**

Note: Bold values indicate the best performance for each metric. G3: Neuropsychiatric (INSOMN, WANDER, DPMOOD, CRYING, ELMOOD); G2: Autonomic nervous (DIZZY, ENERGY, DROWSY, DRYMTH, VISION, FALL); G1: Somatic (NAUSEA, VOMIT, DIARRH, CONSTP, ABDOMN, etc); The AD vs. MCI results in this table are calculated by extracting AD and MCI samples from the three-class COG model.

Table 7
Top EHR features identified by SHAP analysis across Alzheimer's disease classification tasks.

Task	Feature Group	Top Features (Mean |SHAP|)	Clinical Relevance
AD vs. MCI vs. CN	G3 Neuropsychiatric	INSOMN (+0.04), WANDER (+0.03), DPMOOD (+0.03)	ENERGY (+0.07) receives the highest SHAP contribution across stages and may reflect model-level associations with fatigue-related clinical symptoms. Neuropsychiatric symptoms (INSOMN, WANDER, DPMOOD) contribute stably across all disease stages (Kales et al., 2015).
	G2 Autonomic nervous	ENERGY (+0.07), DROWSY (+0.05), DIZZY (+0.03)	
	G1 Somatic	ANKLE (+0.04), MUSCLE (+0.03), URNFRQ (+0.03)	
AD vs. CN	G3 Neuropsychiatric	DPMOOD (+0.16), INSOMN (+0.12), WANDER (+0.07)	Autonomic-related features show prominent contributions in the extreme-contrast task, with DROWSY and ENERGY both reaching +0.30, suggesting that autonomic- and fatigue-related symptoms are strongly used by the model when separating clinically distinct groups. WANDER is consistent with overt goal-directed behavioral deterioration at this stage (da Silva, 2015).
	G2 Autonomic nervous	DROWSY (+0.30), ENERGY (+0.30), DRYMTH (+0.19), FALL (+0.08)	
	G1 Somatic	ANKLE (+0.28), URNFRQ (+0.16), ABDOMN (+0.15)	
AD vs. MCI	G3 Neuropsychiatric	WANDER (+0.11)	The MCI-to-AD transition shows stronger contributions from somatic and autonomic signals. URNFRQ (+0.23) ranks first overall, alongside DIZZY (+0.19) and DROWSY (+0.20), suggesting that autonomic-related symptoms may provide auxiliary model cues for distinguishing AD from MCI (Allan, 2019).
	G2 Autonomic nervous	DROWSY (+0.20), DIZZY (+0.19), ENERGY (+0.10)	
	G1 Somatic	URNFRQ (+0.23), ABDOMN (+0.16), COUGH (+0.16)	
MCI vs. CN	G3 Neuropsychiatric	WANDER (+0.20), INSOMN (+0.09), DPMOOD (+0.06)	The MCI vs. CN task shows notable contributions from sleep-related and behavioral features. WANDER ranks second (+0.20) in this task, suggesting that behavioral symptoms may provide useful model-level cues for distinguishing MCI from CN (da Silva, 2015; Kales et al., 2015).
	G2 Autonomic nervous	DROWSY (+0.24), DRYMTH (+0.09), ENERGY (+0.06), FALL (+0.06)	
	G1 Somatic	URNFRQ (+0.12), PALPIT (+0.10), RASH (+0.08)	

transfer, the probe trained on the source-modality shared representations is directly applied to the corresponding target-modality shared representations without refitting. For the distributional analysis in Table 11, cross-modal distances before and after training are computed using the same preprocessing procedure and the same held-out subjects.

To quantitatively characterize the diagnostic roles of the learned shared and modality-specific subspaces, we conduct quantitative probing in terms of disease predictability, residual utility, and cross-modal transferability, as shown in Table 8. The results show that both shared and modality-specific representations retain disease-discriminative information. Moreover, combining the shared representation with the corresponding modality-specific residual improves the predictive performance over the shared representation alone, indicating that the residual subspaces preserve complementary diagnostic information. The cross-modal transfer results further show that the learned shared representations contain transferable diagnostic information between imaging and EHR.

Table 8
Quantitative probing of learned shared and diagnostic-residual subspaces.

Analysis	Representation	AUC (ADNI)	AUC (NACC)
Disease predictability	$z_{s,\text{img}}$	82.26%	83.64%
	$z_{s,\text{ehr}}$	70.01%	91.78%
	$z_{t,\text{img}}$	83.17%	87.63%
	$z_{t,\text{ehr}}$	68.82%	95.96%
Residual utility	Concat($z_{s,\text{img}}, z_{t,\text{img}}$)	85.32%	85.45%
	Concat($z_{s,\text{ehr}}, z_{t,\text{ehr}}$)	75.58%	96.69%
Cross-modal transfer	z_s: Image→EHR	71.70%	79.83%
	z_s: EHR→Image	68.84%	86.99%

Table 9
Representation-level ablation of shared-specific coding on the ADNI dataset.

Model Variant	Classifier Input	Weighted Voting	AUC (%)
M1	z_s	✗	89.75
M2	Concat($z_{t,img}, z_{t,ehr}$)	✗	88.95
M3	Concat($z_s, z_{t,img}, z_{t,ehr}$)	✗	91.17
M4	**Full Model**	✓	**94.39**

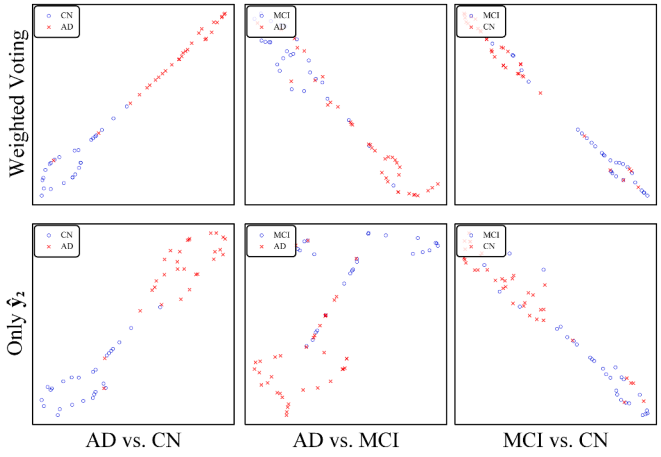


Fig. 7. t-SNE Visualization of Prediction Outputs under Different Decision Strategies. (Weighted fusion: three-branch weighted prediction; Fusion branch: single $\hat{y}_2$ prediction).

We further examine the cross-modal distributional structure of the learned subspaces using RBF-MMD^2, energy distance, and linear-MMD^2. As shown in Table 11, all three measures decrease after training in both shared and modality-specific subspaces. Notably, the discrepancy reduction is consistently larger in the shared subspace, as indicated by the positive ΔG values on both datasets. This suggests stronger cross-modal convergence in the shared representations, while the modality-specific representations retain relatively greater cross-modal discrepancy.

To investigate the contribution of decision-level fusion beyond representation learning, we report the standalone performance of the three prediction branches and compare different decision fusion strategies in Table 13. In addition to the proposed weighted voting, we evaluate simple averaging and majority voting using the same branch predictions. The standalone branches already exhibit substantial diagnostic performance, indicating that the learned branch representations are independently discriminative. Combining the branch predictions further improves overall performance, with weighted voting achieving the highest ACC and AUC on both datasets. These results indicate that weighted voting provides an additional decision-level gain on top of the learned shared-specific representations.

To further validate the effectiveness of each key component in the proposed model, we conducted four groups of ablation experiments on the ADNI and NACC datasets. Specifically, we evaluated the following variants: (1) replacing the image fusion module (DIFFM) with simple feature concatenation, to assess the influence of compact intra-imaging fusion before SCC; (2) removing the multi-branch decision-fusion scheme and retaining only the image branch $\hat{y}_1$ as the final output, to assess the overall contribution of multi-branch decision fusion; (3) removing the tabular modality input and retaining only imaging modalities, to evaluate the complementary role of structured EHR features; and (4) removing CDAB and directly using the original shared-specific representations, to assess the contribution of bidirectional shared-specific attention refinement. The experimental results are shown in Table 10, and the corresponding class-wise prediction results are illustrated in Fig. 8 as bar plots of correct predictions for AD, MCI, and CN classes.

Table 10
Performance comparison under different model settings (%).

Dataset	Setting	ACC	AUC
ADNI	**Full model**	**93.48**	**94.39**
	w/o DIFFM	91.30	92.92
	w/o Multi-branch Decision Fusion	84.78	89.02
	w/o EHR	86.96	88.90
	w/o CDAB	89.13	90.61
NACC	**Full model**	**93.81**	**98.96**
	w/o DIFFM	92.78	97.90
	w/o Multi-branch Decision Fusion	91.75	98.89
	w/o EHR	83.51	92.20
	w/o CDAB	90.72	96.52

Note: Bold values indicate the best performance for each metric.

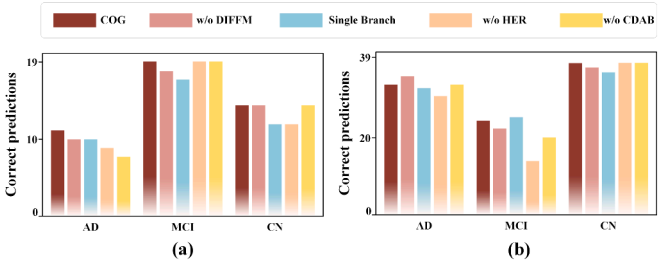


Fig. 8. Correct predictions of AD, MCI, and CN classes across different ablation settings on (a): ADNI and (b): NACC.

As shown in Table 10, replacing DIFFM with simple concatenation leads to a performance drop on both datasets (ADNI: ACC 91.30%, AUC 92.92%; NACC: ACC 92.78%, AUC 97.90%). This result suggests that forming a compact intra-imaging representation before SCC is beneficial for subsequent imaging-EHR shared-specific coding. Compared with simple concatenation, DIFFM provides a more organized imaging-side input for SCC, which helps maintain useful complementary information from different imaging inputs. The class-wise results in Fig. 8 also show that removing DIFFM increases the confusion among disease stages, particularly between AD and MCI.

Removing multi-branch decision fusion and retaining only the image branch $\hat{y}_1$ resulted in a noticeable decline in the classification of MCI cases, suggesting that integrating common, modality-dependent, and interaction-enhanced representations improves the model's robustness across disease stages. When the EHR modality was removed, AUC dropped to 88.90% and 92.20%, highlighting the essential role of structured clinical features, particularly in differentiating MCI from CN when imaging-based cues are less distinctive.

Removing CDAB and using only the initial shared and modality-specific representations leads to performance degradation on both datasets (ADNI: AUC 90.61%; NACC: AUC 96.52%). This shows that shared-specific projection alone cannot fully model the interaction between disease-related semantics and modality-specific diagnostic cues. As a refinement component within SCC, CDAB enables bidirectional

Table 11
Cross-modal distributional discrepancy of shared and modality-specific subspaces before and after training.

Metric	Stage	ADNI			NACC		
		D_{shared}	D_{specific}	ΔG	D_{shared}	D_{specific}	ΔG
RBF-MMD^2	Epoch 0	0.6870	0.6911	**+0.0932**	0.795	0.760	**+0.198**
	Trained model	0.3375	0.4348		0.318	0.481	
Energy distance	Epoch 0	2.0788	2.0492	**+0.3925**	2.565	2.474	**+0.542**
	Trained model	0.9062	1.2691		0.959	1.410	
Linear-MMD^2	Epoch 0	1.6861	1.6849	**+0.2711**	2.031	1.916	**+0.550**
	Trained model	0.9220	1.1919		0.770	1.205	

Note: D_{shared} and D_{specific} denote the cross-modal distributional discrepancies of the shared and modality-specific subspaces, respectively. ΔG denotes the difference between their training-induced discrepancy reductions: $\Delta G = (D^0_{\text{shared}} - D^{\text{trained}}_{\text{shared}}) - (D^0_{\text{specific}} - D^{\text{trained}}_{\text{specific}})$.

Table 12
Ablation study on different loss components on ADNI dataset (%).

Model Variant	Description	ACC	AUC
A	w/o $\mathcal{L}_{CC}$	91.30	90.34
B	w/o $\mathcal{L}_{CLUB}$	89.13	89.76
Ours	**Full Model**	**93.48**	**94.39**

Note: Bold values indicate the best performance for each metric.

Table 13
Comparison of standalone prediction branches and decision-level fusion strategies on ADNI and NACC.

Dataset	Method	ACC (%)	AUC (%)
ADNI	Image branch ($\hat{y}_1$)	84.78	89.02
	Fusion branch ($\hat{y}_2$)	89.13	91.66
	EHR branch ($\hat{y}_3$)	71.74	84.70
	Simple averaging	89.13	90.39
	Majority voting	91.30	91.52
	Ours	**93.48**	**94.39**
NACC	Image branch ($\hat{y}_1$)	91.75	98.89
	Fusion branch ($\hat{y}_2$)	92.78	96.82
	EHR branch ($\hat{y}_3$)	89.69	94.82
	Simple averaging	91.75	96.47
	Majority voting	89.69	95.26
	Ours	**93.81**	**98.96**

Note: $\hat{y}_1$, $\hat{y}_2$, and $\hat{y}_3$ denote the image, fusion, and EHR prediction branches, respectively.

interaction between these representations, enhancing heterogeneous imaging-EHR representation learning.

To verify the contribution of the shared-specific cross-modal coding strategy, we conducted experiments on model variants A and B, as presented in Table 12. Removing $\mathcal{L}_{CC}$ (Variant A) leads to a drop in ACC from 93.48% to 91.30%, suggesting that contrastive consistency helps enhance cross-modal semantic correspondence. Removing $\mathcal{L}_{CLUB}$ (Variant B) results in a further decline to 89.13%, indicating that mutual information regularization helps reduce redundant dependencies between common and modality-dependent representations. These results suggest that both loss components play a complementary role in the overall framework.

4.6. Discussion

4.6.1. Sensitivity analysis of fusion weights

We optimized the branch weights w_1, w_2, and w_3 via grid search on the validation set, with w_2 fixed at 0.5 during the search. Before final prediction, the searched weights are normalized to ensure that their sum equals one. Fig. 9 shows that the best validation performance is obtained when $w_1 = 0.3$ and $w_3 = 0.3$ before normalization. This result indicates that appropriately balancing the fusion branch and modality-specific branches contributes to improved discriminative performance.

Table 14
Overall top-10 discriminative ROIs identified by the model for the COG task on the ADNI dataset.

Rank	ROI	Anatomical Region
1	Thalamus_L	Left thalamus
2	Hippocampus_L	Left hippocampus
3	ParaHippocampal_L	Left parahippocampal gyrus
4	Thalamus_R	Right thalamus
5	Pallidum_L	Left pallidum
6	Hippocampus_R	Right hippocampus
7	Amygdala_L	Left amygdala
8	Caudate_L	Left caudate nucleus
9	Olfactory_L	Left olfactory cortex
10	Pallidum_R	Right pallidum

4.6.2. Visualization of classification results

To demonstrate the interpretability of our proposed method on the COG task, we visualized the attention maps of representative subjects from the AD, MCI, and CN classes for the COG task, as shown in Fig. 10.

In the MRI modality, the most discriminative regions (highlighted in red) are predominantly located in the medial temporal lobe and hippocampus, aligning with established anatomically plausible AD-related saliency patterns. Notably, the attention intensity in the AD subject appears more pronounced than in the MCI and CN subjects, reflecting the progressive nature of structural changes.

Similarly, the PET modality visualization highlights regions associated with metabolic variations. Comparing the two modalities, we observe that while the salient regions in MRI and PET partially overlap (indicating shared pathological semantics), they also focus on distinct areas. This observation suggests that MRI and PET capture complemen-

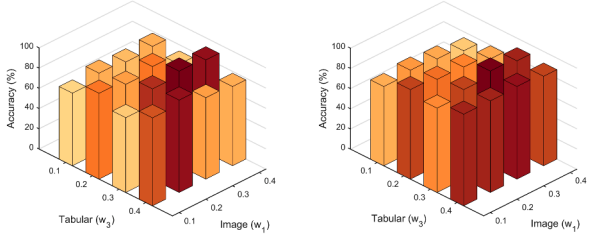


Fig. 9. Accuracy comparisons under different weight distributions. Subplots (a) and (b) report the results on ADNI and NACC datasets.

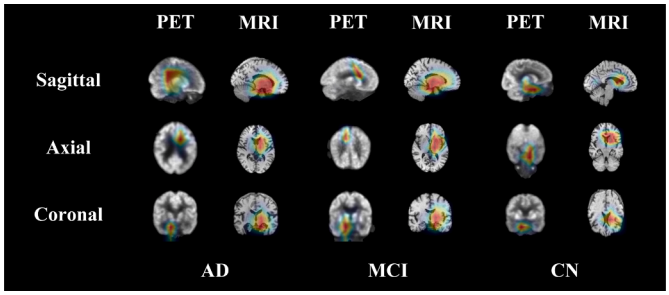


Fig. 10. Visualization of attention maps for MRI and PET on ADNI dataset, with each element denoting the attention maps of a specific brain region.

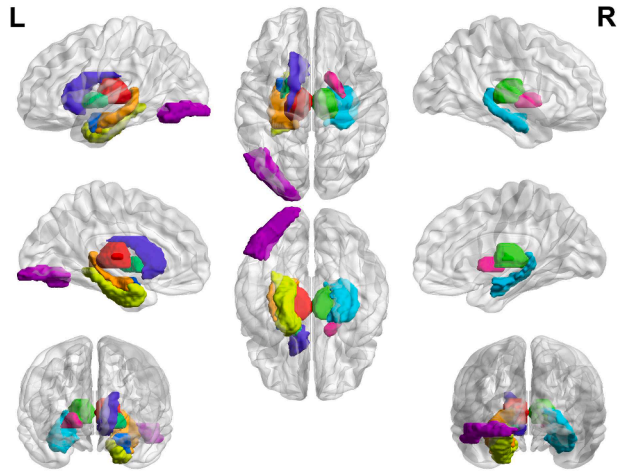


Fig. 11. Visualization of the overall top-10 discriminative ROIs for the COG task on the ADNI dataset.

tary pathological information, which the proposed framework exploits through cross-modal fusion.

To further improve ROI-level interpretability, we used Grad-CAM to generate class activation maps, as summarized in Table 14. The generated activation maps were then mapped to the AAL90 atlas. For each anatomical region, we calculated the average Grad-CAM activation intensity. The top 10 ROIs with the highest overall discriminative scores were selected. As shown in Fig. 11, the core ROIs identified by the model in the COG task were visualized using BrainNet Viewer.

The identified key ROIs mainly included the bilateral thalamus, hippocampus, parahippocampal gyrus, amygdala, olfactory cortex, and basal ganglia-related structures, such as the pallidum and caudate nucleus. These regions are involved in memory consolidation, limbic emotional regulation, olfactory perception, and cortico-subcortical cognitive control. Among them, the hippocampus and parahippocampal gyrus are closely related to episodic memory decline in patients with AD and MCI. Notably, the thalamus and basal ganglia-related regions showed relatively high responses. This may be related to the effective integration of EHR features, which include autonomic, somatic, and neuropsychiatric symptoms. These subcortical structures are involved in autonomic regulation, emotional fluctuation, and behavioral abnormalities. They are also consistent with the clinical phenotypes of AD patients.

5. Conclusion

In this paper, we propose DA-S^{2}RL, a disease-aware shared-specific representation learning framework for heterogeneous image-EHR diagnostic modeling. The core of DA-S^{2}RL is SCC, which organizes heterogeneous multimodal representations into disease-consistent shared semantics and modality-specific diagnostic residuals through cross-modal semantic alignment, redundancy suppression, and bidirectional shared-residual refinement. DIFFM is used as an intra-imaging fusion step to provide a compact imaging representation before SCC. Importantly, the incorporation of structured EHR symptom features is shown to provide task-specific diagnostic information beyond what imaging alone can capture: SHAP-based analysis suggests that neuropsychiatric features contribute most consistently across disease stages, autonomic features are particularly informative at the MCI-to-AD transition, and behavioral symptoms may provide useful model-level cues for distinguishing MCI from CN. The subgroup ablation further demonstrates that incrementally adding clinically grouped EHR features yields consistent performance gains, with the full model achieving accuracies of 93.48% and 93.81% on the ADNI and NACC datasets, respectively. Overall, these results suggest that explicitly organizing heterogeneous multimodal representations according to their shared and residual diagnostic roles can improve image-EHR diagnostic learning beyond direct feature aggregation.

Despite these promising results, several limitations warrant further investigation. The current framework still requires aligned multimodal inputs within each dataset-specific setting, which may not always be available in real-world clinical settings. Future work will explore generative imputation strategies, such as Latent Diffusion Models. Additionally, integrating molecular biomarkers such as APOE genotyping and cerebrospinal fluid markers may further extend the framework's capability toward preclinical AD detection.

CRediT authorship contribution statement

Wei Wang: Writing – original draft, Visualization, Validation, Methodology, Investigation; **Yu Shi:** Validation, Methodology; **Juan Cheng:** Validation, Supervision; **Yu Liu:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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